RAG Embedding Model Comparison

all-MiniLM-L6-v2 (generalist) vs. NeuML/pubmedbert-base-embeddings (biomedical)

Setup

- Same 276 chunks (identical chunking, 4 sources), embedded separately by each model
- 10 queries: 6 direct + 4 paraphrased (no keyword overlap, to test synonym handling)
- results/rag_knowledge_base/ (general) vs. results/rag_knowledge_base_clinical/ (clinical)

Full results (top-1 retrieval)

Query	General: top section (score)	Clinical: top section (score)	Agree?
low left ventricular ejection fraction dilated cardiomyopathy	4.2 Structural abnormalities (from Cine MRI + (0.565)	2.1 Cine MRI (0.590)	No
wide QRS complex bundle branch block	5.3 Bundle Branch Blocks (LBBB / RBBB) (0.428)	5.3 Bundle Branch Blocks (LBBB / RBBB) (0.743)	Yes
Sokolow-Lyon criterion hypertrophy voltage	5.4 Left Ventricular Hypertrophy (LVH) (0.524)	5.4 Left Ventricular Hypertrophy (LVH) (0.588)	Yes
PTB-XL diagnostic superclasses NORM MI STTC CD HYP	(document start) (0.530)	(document start) (0.526)	Yes
ACDC dilated cardiomyopathy hypertrophic cardiomyopathy classification	4.2 Structural abnormalities (from Cine MRI + (0.601)	4.2 Structural abnormalities (from Cine MRI + (0.566)	Yes
ECG preprocessing baseline wander wavelet denoising	2. The few multimodal approaches that exist o (0.671)	4.4 Inflammatory / infiltrative disease (from (0.724)	No
reduced pumping function of the heart chamber	1.2 Why MRI for the heart specifically ? (0.494)	2.1 Cine MRI (0.557)	No
irregular heartbeat coming from upper chambers	5.1 Atrial Fibrillation (AF) (0.470)	3.1 The Waves (0.563)	No
heart attack scar tissue detection	4.3 Tissue/scar abnormalities (from LGE) (0.582)	2.2 LGE (Late Gadolinium Enhancement) (0.654)	No
thickened heart muscle wall abnormal	4.2 Structural abnormalities (from Cine MRI + (0.612)	4.2 Structural abnormalities (from Cine MRI + (0.627)	Yes

Honest analysis

- 6/10: both models agree - clinical model scores higher confidence on the same answer (e.g. 0.428 vs. 0.743)
- 4/10: models disagree
- **2 clear clinical-model failures**, both on paraphrase queries: "ECG preprocessing..." -> general correctly found the real source (0.671), clinical found an unrelated section (0.724, HIGHER confidence, WRONG)
- Same pattern on "irregular heartbeat..." (AFib): general correct (0.470), clinical wrong (0.563)
- 2/10: both defensible, different granularity - not a real disagreement
- **Key concern**: clinical model's wrong answers scored HIGHER confidence than when correct - harder to catch downstream

Round 1 - Pearson correlation (easy negatives)

- Methodology from NeuML's own model card: correlate similarity score against known relevance
- 10 queries x (1 correct chunk + 3 clearly-unrelated chunks) = 40 labeled pairs

Model	Pearson r	Mean sim (relevant)	Mean sim (irrelevant)
General	0.601	0.468	0.246
Clinical	0.678	0.512	0.264

- Clinical model wins here (both $p < 0.0001$) - but easy negatives can't detect the failure mode found above

Round 2 - harder Pearson test (25 queries, hard negatives)

- Expanded to 25 queries (10 from round 1 + 15 new), covering all clean sections of the source document
- Added 2 **hard negatives** per query: another query's correct answer - plausible but wrong
- 150 pairs total: 25 positive + 50 hard negative + 75 easy negative

The 25 queries and their ground-truth (ok) section:

#	Query	Ground-truth section
1	low left ventricular ejection fraction dilated cardiomyopathy	4.2 Structural abnormalities
2	wide QRS complex bundle branch block	5.3 Bundle Branch Blocks
3	Sokolow-Lyon criterion hypertrophy voltage	5.4 Left Ventricular Hypertrophy
4	PTB-XL diagnostic superclasses NORM MI STTC CD HYP	(document start)
5	ACDC dilated cardiomyopathy hypertrophic cardiomyopathy classification	4.2 Structural abnormalities
6	ECG preprocessing baseline wander wavelet denoising	2. The few multimodal approaches
7	reduced pumping function of the heart chamber	2.1 Cine MRI
8	irregular heartbeat coming from upper chambers	5.1 Atrial Fibrillation
9	heart attack scar tissue detection	4.3 Tissue/scar abnormalities
10	thickened heart muscle wall abnormal	4.2 Structural abnormalities
11	P wave QRS complex T wave cardiac cycle sequence (new)	3.1 The Waves
12	RR interval PR interval QT interval definitions (new)	3.2 The Intervals
13	ST segment elevation myocardial infarction STEMI leads (new)	5.2 ST-Elevation Myocardial Infarction
14	regional heart wall motion hypokinetic akinetic dyskinetic (new)	4.1 Functional abnormalities
15	myocarditis inflammation T2 signal elevated (new)	4.4 Inflammatory / infiltrative disease
16	T1 T2 mapping tissue characterisation (new)	2.3 T1 / T2 mapping
17	cardiac MRI sequences overview Cine LGE mapping (new)	2. Cardiac MRI sequence
18	why is MRI used for the heart (new)	1.2 Why MRI for the heart specifically
19	how electrical signals recorded skin electrodes leads (new)	1.1 How Are Signals Recorded
20	twelve lead ECG sampling rate data shape (new)	1.3 Data Dimensions (Standard ECG)
21	short axis stack slices base to apex 3D reconstruction (new)	2.4 Short axis stack
22	end diastolic volume blood filling ventricle at rest (new)	2. End-diastolic volume (EDV)
23	ejection fraction percentage blood pumped per beat (new)	1. Ejection Fraction (EF)
24	feature extraction morphology temporal spatial levels ECG (new)	4. Feature Extraction, Three Levels
25	what is an electrocardiogram basic definition (new)	1. What is an ECG?

Model	Pearson r (all 150)	Pearson r (positive vs. hard negatives, 75)
General	0.693	0.796
Clinical	0.716	0.773

- Full mix: clinical still wins (0.716 vs. 0.693) - coarse separation, consistent with round 1
- **Hard negatives isolated: general model wins (0.796 vs. 0.773)** - this is the task RAG retrieval actually performs
- Triangulates with the top-1 test's 2 clinical-model failures, via an independent, larger method

Final conclusion

- **Keep all-MiniLM-L6-v2 in production** - firm conclusion, not a close call
- Two independent tests agree: general model is more reliable at picking the correct section among plausible candidates
- Clinical model's advantage (coarse relevant/irrelevant separation) doesn't transfer to this harder, decision-relevant task
- Real cost already observed: 5x larger download, extended network instability during setup
- No case for switching